

SMART INDIA HACKATHON 2026 • OFFICIAL TECHNICAL BLUEPRINT

YATRISATHI — FRONTEND WORKFLOW SPECIFICATION

Smart Tourist Safety Monitoring & Incident Response System • SIH 2026 Core

Section 1: Technology Stack, File Locations & Implementation Roles

Structured catalog of core client libraries, target source files, and concrete implementation responsibilities across the application ecosystem.

Technology Stack	File Locations	Implementation Roles
JavaScript (ES6+) / JSX	<code>client/src/**/*.jsx</code>	Component UI state (useState), event triggers, modal dispatch, and localStorage transaction logic.
HTML5	<code>index.html</code> , semantic DOM trees	Base layout scaffolding, accessible modal dialogs, and native hardware hooks (href="tel:112").
Tailwind CSS & CSS3	Component <code>className</code> tags	Responsive safety interface, slide-in drawer animations, verified/active badges, and emergency red SOS styling.
React 18	<code>client/src</code> component ecosystem	Virtual DOM updates, single-page application (SPA) architecture, component reusability, and popover lifecycles.
Lucide React	<code>NavigationDrawer.jsx</code> , <code>Modal headers</code>	Scalable, high-contrast SVG vector icons (Home, ShieldCheck, WifiOff, Info, FileText, PhoneCall).
Vite	<code>vite.config.js</code> , root dev pipeline	High-speed Hot Module Replacement (HMR) local dev server (localhost:5173) and optimized static production builds.
Web Storage (LocalStorage API)	<code>ProfileModal.jsx</code> , <code>telemetry utils</code>	Zero-latency client persistence for SafeTrip ID (YS-OD-XXXX), tourist medical notes, blood group, and emergency kin contacts.
HTML5 Geolocation API	<code>MapContainer.jsx</code> , <code>telemetry services</code>	Live GPS coordinate polling (latitude/longitude) for real-time tracking, incident dispatch, and geofence boundary checks.

Section 2: End-to-End Frontend Workflow Tree

Hierarchical view of component mounting, runtime triggers, state transitions, and persistent storage lifecycles.

```
yatrisathi-frontend-architecture - tree-view (v1.0.0)

YATRISATHI FRONTEND CLIENT (index.html -> main.jsx)
├── [Global Shell] App.jsx
│   ├── Top Header (YS Icon Badge, App Title, Drawer Trigger Toggle)
│   ├── [Core Viewport] MapContainer.jsx
│   │   ├── Leaflet Interactive Map View (Hirakud-Sambalpur Mesh Region)
│   │   ├── Real-time GPS Position Marker (navigator.geolocation)
│   │   └── Safe vs Restricted Tourist Geofence Polygons
│   └── [Slide-Over Panel] NavigationDrawer.jsx
│       ├── 1. Home (Status: ACTIVE)
│       │   └── Action: Closes drawer and recenters viewport to tourist's live position.
│       ├── 2. Profile (Status: VERIFIED)
│       │   ├── Action: Triggers isProfileOpen -> mounts ProfileModal.jsx
│       │   ├── Displays read-only SafeTrip ID (YS-0D-XXXX)
│       │   ├── Editable inputs: Name, Kin Phone, Blood Group, Stay Info
│       │   └── Storage Sync: Commits JSON payload to localStorage ('yatrisathi_profile')
│       ├── 3. Offline Sync & Settings (Status: CACHE ARMED)
│       │   └── Action: Inspects service worker caching and pending telemetry buffer queue.
│       ├── 4. About (System Info)
│       │   ├── Action: Triggers isAboutOpen -> mounts AboutModal.jsx
│       │   ├── Displays SIH 2026 Core version (v1.0.0)
│       │   └── Outlines incident classification engine and edge telemetry protocols.
│       ├── 5. Privacy & Legal
│       │   └── Action: Renders DPDP compliance standards and tourist data anonymity policy.
│       └── 6. Help / 112 Helpline (Red SOS Card)
│           └── Action: Directly triggers hardware phone dialer: tel:112
└── [Background Layer] Offline Incident Queue
    ├── Connectivity State Listener (window.ononline / window.onoffline)
    └── Local storage telemetry buffer queued for upstream server synchronization
```

Section 3: Incident Dispatch & SOS Pipeline Summary

Step 1 (Activation) Tap 'Help / 112 Helpline' to trigger instant emergency calling.	Step 2 (Telemetry Extraction) Captures device coordinates and battery state in background.	Step 3 (Offline Fallback) If cellular data is dropped, telemetry records are queued in local cache for delayed sync upon network recovery.
---	--	--